

CSMMECH000000003 Phantom Menahony Zookeeper Proof of Concept

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** CSMMECH000000003 Phantom Menahony Zookeeper Proof of Concept V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
Piezoelectric material	V1.0: PZT-5H; d33=289 pC/N; Tc=193°C; lead-containing	V2.0: KNbO3-BaTiO3; d33=285 pC/N; Tc=310°C; lead-free	RoHS/REACH compliance — PZT restricted; Tc +117°C prevents thermal depolarization	TDK 2025 lead-free catalog; J. Am. Ceram. Soc. 108 (2025)
Post-event navigation	V1.0: GPS assumed operational	V2.0: LoRa mesh 915 MHz, 256-node, 2-5 km/hop; MRAM offline maps	GPS vulnerable to solar energetic particle damage in Carrington scenario	ITU-R WP 4C 2024; LoRa Alliance v1.1

Parameter	V1.0	V2.0	Reason	Primary Source
Composite matrix — recyclability	V1.0: Thermoset epoxy; irreversible crosslink; landfill end-of-life	V2.0: Elium thermoplastic acrylic; thermal solvolysis 350°C; 95% fiber + 100% monomer recovery	Thermoset blocks Phoenix Protocol circular economy mandate	Arkema Elium TDS 2025; Composites Part A 2024

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
Piezoelectric material	Lead restricted under RoHS Annex II; Tc 117°C lower = premature depolarization risk	HIGH
Post-event navigation	GPS invalid assumption in Carrington scenario; navigation fails in primary use case	HIGH
Composite matrix — recyclability	No end-of-life recovery path; contradicts Phoenix Protocol; Phoenix Protocol breach	HIGH

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMMECH000000003 Phantom Menahony Zookeeper Proof of Concept V1.0 archived; V2.0 is the active specification as of June 2026.